

DAKDL Baby-sitting Service Case-Study

Business Context

The DAKDL Business Association operates a babysitting service for the general public. The association has branches (Clubs) distributed throughout the country. When customers contact them, coordination staff collect customer information including name, address, and phone number. The coordination staff then assign babysitting jobs to the Club nearest to the customer's location and assign available employees for the requested date. The coordination staff record details for each babysitting job, including the amount paid by the customer and the employee responsible for that job.

Objective

Design and implement a Data Warehouse (Star Schema) and an ETL process to support the analytical needs of the DAKDL DAKDL Baby-sitting Service.

Business Requirements

The Chief Financial Officer of DAKDL Business Association wants to create a data warehouse to analyze the association's operations. The CFO needs to know:

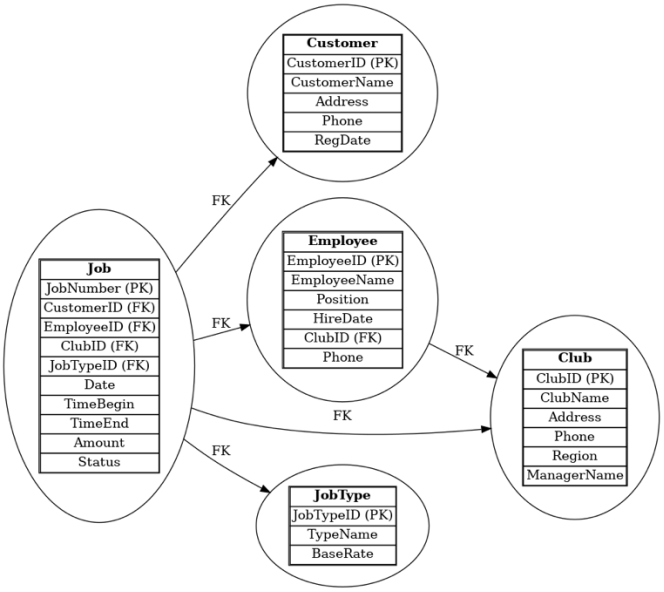
- Amount paid by each customer per job, weekly, monthly, or yearly
- Total revenue generated by each employee for the association, weekly, monthly, or yearly
- Total revenue generated by each Club for the association, weekly, monthly, or yearly
- Working hours of each employee per job, during holidays, weekly, monthly, or yearly
- Employee working hours report during holidays
- Revenue report by Club
- Quarterly trend reports

Briefly The CFO requires a system that can answer the following analytical questions:

- **Revenue Analysis:** Amount paid by each customer per job, weekly, monthly, or yearly.
- **Performance Analysis:** Total revenue generated by each employee and each Club (weekly, monthly, or yearly).
- **Labor Analysis:** Working hours of each employee per job, including holiday tracking.
- **Trend Reporting:** Quarterly trend reports and revenue reports by Club.

Source System

The UML diagram below shows the OLTP system of DAKDL Business Association:



Assignment

Task 1: Source System (OLTP) Setup

Option 1: CSV-Based Source

Provide CSV files (Club.csv, Customer.csv, Employee.csv, JobType.csv, and Job.csv) as your raw data source. You must extract your data directly from these files during the ETL process.

Option 2: SQL-Based Source

Based on the provided OLTP diagram:

1. **Create the Database:** Use SQL to create the following tables: Club, Customer, Employee, JobType, and Job.
2. **Define Constraints:** Ensure all Primary Keys (PK) and Foreign Keys (FK) are correctly implemented as shown in the source schema.
3. **Populate Data:** Insert at least **5–10 sample records** into each table to ensure you have data to "extract" later.
    - ***Note**:* Ensure at least one Job date falls on a holiday (e.g., New Year or Songkran) to test the CFO's holiday reporting requirement.

Task 2: The 4-Step Design Process

1. **Identify the Business Process:** What is the specific activity being measured?
2. **Declare the Grain:** Define exactly what one row in the fact table represents.
3. **Identify the Dimensions:** List the tables that provide context (Who, Where, When, What).
4. **Identify the Facts:** List the quantitative measures (e.g., Revenue, Duration).

Task 3: Star Schema Diagram

Draw a full **Star Schema Diagram**. You must use professional tools such as **Draw.io, PowerPoint, or Lucidchart**. Ensure all primary keys (PK), foreign keys (FK), and attributes are clearly labeled.

Task 4: Physical Schema Creation

Generate the **SQL DDL** scripts to create your Star Schema. This includes:

- CREATE TABLE statements for all Dimension tables.
- CREATE TABLE statements for the Fact table, ensuring appropriate data types and constraints.

Task 5: ETL Implementation

Develop the ETL process to populate your Star Schema. You may use **RapidMiner, SQL, or Python (Pandas)**. Your process must:

1. **Load Dimensions:** Extract data from OLTP and load into DWH dimensions (include a Date dimension).
2. **Load Fact Table:** Extract job records, perform necessary transformations (e.g., calculating "Working Hours" from TimeBegin and TimeEnd), and map keys to the fact table.

Task 5: Data Deployment & Validation

Run **5 SQL Queries** against your populated Star Schema to prove it meets the CFO's requirements:

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1. Total revenue per Club per Month.
2. Top 5 employees by revenue generated in a specific year.
3. Average working hours of employees per job type.
4. Total amount paid by a specific customer quarterly.
5. A report showing employee hours worked specifically during holiday dates.

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5. Submission Guidelines

Submit a single ZIP file containing:

- **Design Document:** A PDF including the 4-step process and the Star Schema diagram.
- **SQL Scripts:** The DDL for table creation and the 5 validation queries.
- **ETL Files:** Your RapidMiner process (.xml), Python script (.ipynb), or SQL ETL script.
- **Dataset:** A small sample of the populated data (CSV or SQL dump) for the TA to verify.